

COLLEGE RECOMMENDATION SYSTEM USING PROLOG

Abstract

Choosing an appropriate college is one of the most important decisions for students after completing their secondary education. Students often consider multiple factors such as entrance examination scores, college rankings, placement statistics, fees, scholarships, location preferences, hostel facilities, and course availability. Analyzing all these parameters manually can be difficult and time-consuming.

This project presents a rule-based College Recommendation System developed using Prolog. The system uses logical inference and expert system principles to recommend colleges according to a student's academic profile and preferences. The recommendation process incorporates eligibility checking, branch availability, budget constraints, hostel requirements, placement opportunities, accreditation quality, and location preferences.

The system demonstrates the use of knowledge representation, logical reasoning, recursion, list processing, and expert system design in Prolog.

1. Introduction

College selection is a multi-criteria decision-making problem. Students often rely on rankings or opinions without considering all relevant factors simultaneously. A recommendation system can assist students by evaluating multiple parameters and providing personalized suggestions.

Prolog is particularly suitable for such applications because it supports:

- Knowledge representation
- Logical inference
- Rule-based reasoning
- Backtracking
- Expert system development

The College Recommendation System stores information about colleges and uses logical rules to recommend suitable institutions for students.

2. Problem Statement

Students face several challenges while selecting colleges:

- Large number of available institutions
- Different admission criteria
- Variation in fees and scholarships
- Differences in placement opportunities
- Hostel and infrastructure considerations
- Location preferences

The objective is to build a rule-based expert system that automatically recommends colleges based on student requirements.

3. Objectives

The primary objectives of this project are:

- To develop a college recommendation expert system.
 - To represent college information using Prolog facts.
 - To implement eligibility checking using admission cutoffs.
 - To recommend colleges based on student preferences.
 - To compare colleges using multiple parameters.
 - To classify colleges into tiers.
 - To provide scholarship and placement insights.
 - To demonstrate logical inference and reasoning.
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4. Technologies Used

Technology	Purpose
Prolog	Logic Programming
SWI-Prolog	Development Environment
VS Code	Code Editor
GitHub	Version Control and Project Hosting

5. System Architecture

The system follows a rule-based expert system architecture.

Student Input

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Eligibility Checking

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Branch Verification

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Budget Filtering

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Hostel Verification

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Scoring Engine

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Recommendation Engine

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Recommended Colleges

The recommendation engine evaluates colleges using predefined facts and logical rules.

6. Knowledge Base Design

The knowledge base stores information about colleges using Prolog facts.

College Information

- College Name
- Location
- Type
- Ranking
- Fees
- Accreditation
- Seats

Example:

```
college_name(iit_bombay,'IIT Bombay').  
college_rank(iit_bombay,3).  
college_fee(iit_bombay,200).
```

Courses Offered

```
course_offered(iit_bombay,btech,computer_science).
```

Admission Cutoffs

```
admission_cutoff(nit_trichy,jee_main,97,100).
```

Placement Statistics

```
placement_stat(iit_bombay,28,300,95).
```

Scholarship Information

```
scholarship(iit_bombay,both,100).
```

Hostel Availability

```
hostel_available(iit_bombay,yes).
```

7. Rule Base Design

The system uses logical rules to infer recommendations.

Important rules include:

Eligibility Checking

`eligible(CollegeID,ExamType,Score).`

Determines whether a student satisfies admission requirements.

Branch Availability

`branch_available(CollegeID,Course,Branch).`

Checks if a branch is offered.

Fee Affordability

`fee_affordable(CollegeID,Budget).`

Ensures the college is within budget.

Hostel Verification

`hostel_ok(CollegeID,NeedHostel).`

Checks hostel availability.

8. Recommendation Engine

The recommendation engine applies filtering and scoring mechanisms.

Filtering Stage

The following constraints are checked:

- Admission eligibility
- Branch availability
- Budget constraints
- Hostel requirement

Scoring Stage

Colleges are evaluated using:

- Ranking Score
- Placement Score
- Fee Score
- Accreditation Score
- Location Preference Score

Total Score Formula

Total Score = Rank Score + Placement Score + Fee Score + Accreditation Score
+ Location Score

The colleges are then sorted according to their scores.

9. Additional Features

Tier Classification

Colleges are classified into:

- Tier 1
- Tier 2
- Tier 3
- Tier 4

based on ranking.

Scholarship Advisory

Identifies colleges where students may receive scholarships.

Dream Colleges

Suggests highly ranked colleges where the student is eligible.

Safe Colleges

Suggests colleges where the student comfortably exceeds admission cutoffs.

Stretch Colleges

Suggests colleges that are slightly above the student's current eligibility level.

College Comparison

Allows comparison of:

- Rankings
- Placements
- Fees

between two colleges.\

10. Sample Queries

View College Profile

?- college_profile(iit_bombay).

Compare Two Colleges

?- compare_colleges(vit_vellore,srm_chennai).

Check Eligibility

?- eligible(nit_trichy,jee_main,98).

Generate Recommendations

?- recommend_top5(
 jee_main,
 96,
 btech,
 computer_science,
 300,
 delhi,
 yes).

11. Results

The system successfully:

- Stores college information.
- Performs logical inference.
- Checks admission eligibility.
- Filters colleges according to constraints.
- Generates personalized recommendations.
- Compares institutions.
- Provides scholarship and placement insights.

The recommendation engine produces ranked college suggestions according to user requirements.

12. Advantages

- Fast decision-making
- Personalized recommendations
- Logical and explainable reasoning
- Easy to maintain and extend

- Demonstrates expert system concepts
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13. Limitations

- Uses static college data.
 - Does not access real-time rankings.
 - Limited to predefined colleges.
 - No graphical user interface.
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14. Future Scope

Future improvements may include:

- Web-based deployment
 - Graphical user interface
 - Database integration
 - Real-time ranking updates
 - Machine learning-based recommendation system
 - Integration with counseling platforms
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15. Conclusion

The College Recommendation System demonstrates the application of Prolog in building a rule-based expert system. The project effectively combines knowledge representation, logical reasoning, recursion, and decision-making techniques to recommend colleges based on student preferences. The system provides a practical example of how expert systems can assist students in making informed educational decisions.

References

1. Bratko, I. Prolog Programming for Artificial Intelligence.
2. SWI-Prolog Official Documentation.
3. Russell, S. and Norvig, P. Artificial Intelligence: A Modern Approach.
4. NIRF Ranking Framework.
5. NAAC Accreditation Guidelines.